



An automatic feature point extraction method based on laser vision for robotic multi-layer multi-pass weld seam tracking

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Abstract

Laser vision-based seam tracking has been an important research hotspot in modern welding manufacturing. However, severe noise interference during welding and the complex contour curves of filling welds hinder the development of high-precision seam tracking in multi-layer multi-pass (MLMP) welding. To solve this problem, a point distribution model (PDM) has been implemented to express the laser stripe pattern of MLMP welds. Then, an end-to-end feature point extraction algorithm is proposed. The “coarse-to-fine” positioning strategy achieves global correlation and local constraints. The low-resolution heatmap regression and coordinate offset regression balance the efficiency and precision, where the backbone is improved with attention mechanisms. Furthermore, the soft coordinate loss and the Gaussian mixture model were combined to improve the generalization performance. Based on the model, an automatic ROI extraction method and output points filtering are implemented to complete the whole tracking process. In experiments, the proposed method achieved good tracking performance even under strong noise, with the mean absolute error (MAE) being controlled within 0.3 mm. The feature point extraction method shows advantages in both precision and stability, laying a foundation for advanced robotic MLMP welding production.

Keywords Multi-layer multi-pass welding · Laser vision sensor · Point distribution model · Active shape model · Deep learning · Seam tracking

1 Introduction

With the continuous development of the modern manufacturing industry, to achieve reliable connection of medium- and thick-plate structural components, multi-layer and multi-pass (MLMP) welding methods are commonly used. This welding method not only achieves low heat input welding through multiple weld bead filling but also brings metallurgical advantages related to thermal cycling and structural

segregation to the welded joint [1]. At present, robotic automatic welding has become an important part of advanced manufacturing technology. This ensures the stability of the welding process and improves the efficiency of thick-plate welding. However, due to factors such as processing errors, fluctuations in assembly gaps, and random thermal deformation of joints, it is hard to achieve high-precision MLMP welding based on the current “teaching and play-back” mode. Therefore, to boost the development of the robotic intelligent welding process, the laser vision sensing technique has been widely used. The seam tracking technology based on laser vision has become a research hotspot at home and abroad.

At present, there are two main methods for seam tracking based on laser vision sensing: the detector-based method and the tracker-based method [2]. The former often uses mathematical methods to fit laser stripes and locate weld seam feature points through their geometric features. This method usually contains two steps: stripe extraction and feature point extraction. Wang et al. [3] proposed an iterative adaptive algorithm and combined it with the random sample and consensus (RANSAC) algorithm to fit laser stripe points. It solved the target tracking point position through

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the fitting equation. The tracker-based methods are mostly derived from correlation filtering methods, which exhibit better tracking performance in terms of accuracy and speed. Zou et al. [4] proposed using the spatiotemporal context (STC) model combined with the traditional morphological method. Zhou et al. [5] used an efficient convolutional operator ECO to achieve seam tracking for different joint forms, further improving the accuracy with the help of particle filtering. However, the problem lies in the tendency to cause tracking drift with the interference of strong welding noise. Moreover, it often requires setting an initial tracking point for subsequent tracking.

In addition, during the seam tracking process, the appearance of welding spatter, arc light, and reflections usually affects the stability and accuracy of the tracking methods. To deal with this, studies have been made to reduce their interference. Traditional methods include using directional filtering and inter-frame minimization. With the continuous development of deep learning for various visual tasks, scholars have also begun to apply it to laser stripe extraction. Zou et al. [6] used a deep network based on a conditional generative adversarial network (CGAN) for generating laser stripe parts disturbed by noise, laying the foundation for subsequent weld feature point recognition. Wang et al. [7] designed an algorithm for laser stripe segmentation based on VGGNet. However, the introduction of complex networks may affect overall efficiency. Some researchers increased the anti-noise ability by modeling the characteristics of the laser stripes. He et al. [8] proposed the use of a visual attention model to enhance the saliency of weld contour for laser stripe extraction and feature point detection under the interference of arc light from the molten pool. For corner joints, Chen et al. [9] designed a specific convolutional template based on stripe features for likelihood calculation, achieving strong noise resistance. The idea of focusing on laser stripe features may better adapt to the actual welding tracking process.

However, these seam tracking methods still have certain limitations. Firstly, although research has been made on weld seam tracking methods for various joint forms and groove types, most of them focus on thin-plate welding or the root pass of thick-plate welding with obvious characteristics. The subsequent filling of MLMP weld has a more urgent need for seam tracking due to issues such as thermal deformation accumulation. The complex composition of surface contour curves for MLMP welds also poses challenges for feature point extraction. Xiao et al. [10] used an algorithm based on an improved snake model to extract laser stripes under noisy conditions and obtained the position of actual weld bead feature points through the curvature method. He et al. [11] extracted weld contour by constructing SIFT features and directional detection, and further utilized Bayesian network models to make inference decisions on multiple feature points obtained. At present, there are few studies applied to

robotic MLMP seam tracking, which often requires multi-step extraction methods.

To solve the above problems, there is an urgent need for a method that avoids environmental noise interference by focusing on the characteristics of the laser stripe. It is necessary to be able to adapt to the tracking process of different passes of MLMP welds. Zhao et al. [12] designed a tracking method based on EFRNet, which can extract laser stripes under strong noise and obtain seam feature points through regression. Yang et al. [13] proposed an end-to-end method to detect weld groove edge points under strong noise. What is worth mentioning is that similar coordinate regression methods have been extensively studied for other visual tasks, such as face detection [14] and human pose estimation [15]. Through experiments, it can be found that these visual tasks share certain similarities and commonalities with MLMP weld feature point extraction. This solution for locating landmark points from images of deformable objects under noise conditions [16] can be transferred and referenced. The shape distribution of weld stripes is relatively stable, which has the potential to achieve accurate weld feature point localization.

Therefore, inspired by the above research, this paper proposes a robotic MLMP weld seam tracking method based on a point distribution model (PDM). The main contributions of this method are threefold. Firstly, based on stripe features, a feature PDM is proposed to model the MLMP weld laser stripes effectively. Secondly, an end-to-end laser stripe feature point tracking network (LSFPTN) was designed, with a complete process for MLMP seam tracking. Third, the proposed method utilizes the PDM distribution characteristics to locate the target point even under strong noise interference.

The rest of this paper is arranged as follows. Section 2 introduces the robotic MLMP welding system based on a laser vision sensor. Section 3 shows the establishment and evaluation of an MLMP weld laser stripe feature PDM. The structure and loss function composition of the LSFPTN network were discussed in detail. In Section 4, automatic ROI extraction and feature point filtering are proposed. Section 5 provides a detailed demonstration of the experimental process for validation of the proposed tracking method for MLMP welding. Finally, Section 6 summarizes this paper.

2 Laser vision-based MLMP weld seam tracking system

Figure 1 shows the laser vision-based robotic MLMP weld seam tracking system. The system mainly consists of three parts: a welding robot and its control cabinet, a self-developed laser vision sensor, and an image processing and communication module. The laser vision sensor is placed on the end of the robot welding gun. The camera of the sensor is connected to the industrial computer through a network cable to achieve the

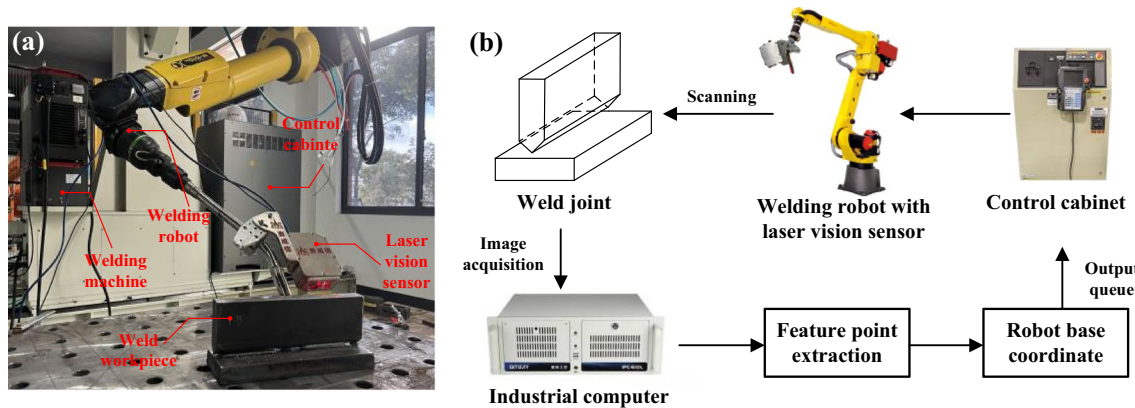


Fig. 1 The **a** picture and **b** diagram of the laser vision-based robotic weld seam tracking system

transmission of laser images. The welding robot communicates with the control cabinet through TCP/IP protocol. It adjusts the position of the welding torch end after receiving the three-dimensional robot base coordinate sent by the control cabinet.

The laser vision sensor consists of a single-line laser transmitter and a CCD camera. A narrow band bandpass filter of 650 ± 10 nm was installed in front of the camera to further reduce the interference of external light sources on weld seam recognition. The camera captures laser weld seam images at a frequency of approximately 20 Hz. The geometric optical model of laser sensing is shown in Fig. 2. Based on the laser triangulation principle, the 3D coordinates of welding seam feature points can be solved through the calibration matrix.

To obtain the three-dimensional coordinates of feature points through the laser vision sensing system, it is necessary to convert the two-dimensional point coordinates in the laser stripe image. Based on our previous works [17], the calibration process includes camera calibration, laser plane calibration, and hand-eye calibration. Among them, the internal parameter matrix of the camera is obtained through camera calibration. Laser plane calibration achieves the conversion from laser plane coordinates to camera coordinate systems. The hand-eye calibration calculates the transformation matrix between the robot tool

coordinate system and the camera coordinate system. The coordinate conversion formula is as follows:

$$P_b = T_b^t * T_t^c * T_c^l * P_l \quad (1)$$

Among them, P_l is the homogeneous form of pixel coordinates in the image coordinate system; T_c^l is the transformation matrix from the laser plane coordinate system to the camera coordinate system; T_t^c is the transformation matrix from camera coordinates to tool coordinate systems; T_b^t is the transformation matrix from the tool coordinate system to the robot coordinate system, given by the specific pose of the robot; P_b is the homogeneous coordinate of the point in the robot base coordinate system. The formulas for the main transformation matrix are shown in Table 1.

3 PDM-based MLMP feature point extraction

3.1 Feature point distribution model

As mentioned above, the main challenges of MLMP weld seam tracking mainly lie in two aspects: severe welding noise

Fig. 2 The laser vision sensor: **a** the composition and **b** the model of the structured laser vision system

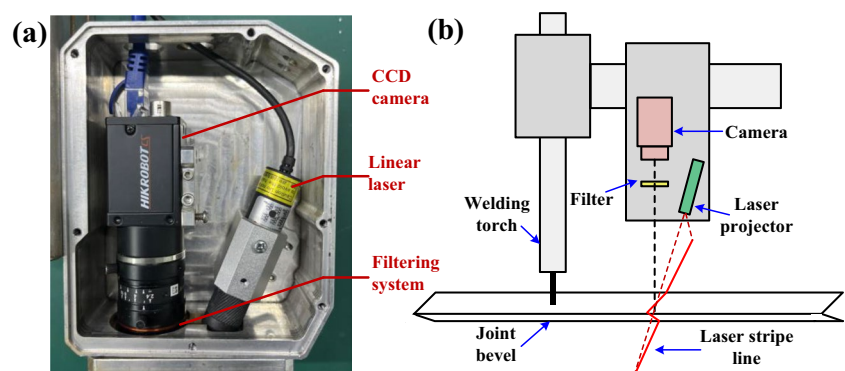
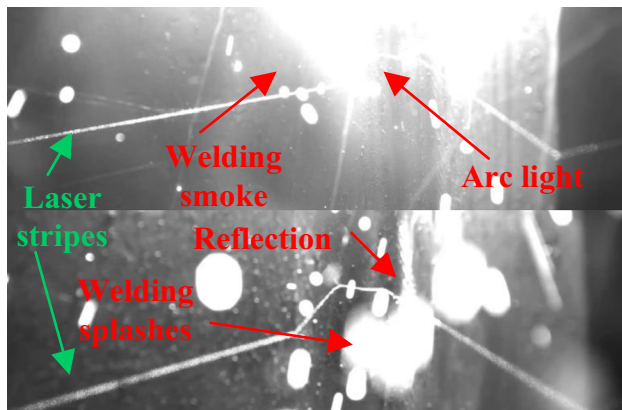


Table 1 Transformation matrixes of the laser vision model

Parameters	Formula
Camera calibration matrix	$M_{camera} = \begin{bmatrix} \alpha_x & \gamma & u_0 \\ 0 & \alpha_y & v_0 \\ 0 & 0 & 1 \end{bmatrix}$
Hand-eye transformation matrix	$T_c^r = \begin{bmatrix} {}^C R & {}^C T \\ 0_{3 \times 1} & 1 \end{bmatrix}$

**Fig. 3** The demonstration of interference of welding noise for MLMP seam tracking

interference (as shown in Fig. 3) and difficulty in extracting feature points from complex curves of MLMP welds (as shown in Table 2). The MLMP laser stripes are different from the simple morphology of thin-plate weld or root weld beads. The laser stripes that are projected onto MLMP weld beads exhibit different morphological features. Also, the deformation influenced by random factors brings in more changes. Although the complex curve features are not clear, the feature point distribution of the laser stripe shows a certain pattern.

To use this feature point distribution pattern as a prior condition for accurate seam feature point extraction, the PDM is established. To better represent the stripe shapes, the points that can more accurately express the stripe morphology are selected as the landmark points. The most important feature points in the shape of MLMP stripes are the inflection points at each position. Therefore, through the second-order derivative of the fringe points, four points can be extracted as a component of the PDM, namely, two groove position points and two feature points at the weld bottom to be filled. The calculation schematic is shown in Fig. 4. To further strengthen the local constraints of the distribution model, three auxiliary localization feature points are inserted in the middle of the four selected feature points along the laser stripe. Then two auxiliary feature points outward at the edge plate position are added. Finally, the PDM consisting of nine ordered landmark points was

formed. Table 2 shows the distribution model of key weld feature points in laser stripes in different layers.

3.2 Discussion on the effectiveness of PDM

Based on the above definition, the PDM captures the non-rigid deformation of objects. The basic principle of feature point localization is based on the cooperation of position relations and response functions of the two nearest landmark points. Derived from this, the active shape model (ASM) and active appearance model (AAM) [18] were proposed.

As a widely used feature point extraction method for pattern recognition, AAM combines statistical analysis based on local features and information on global shape and texture in the modeling process. The landmark positioning can be summarized as an iterative process of “matching, comparison, and rematching.” Based on this traditional method, experiments were conducted to verify the effectiveness of the proposed laser stripe feature PDM.

During the matching process, principal component analysis (PCA) is used to describe the dynamic changes of feature points. The current matching performance is evaluated based on image gradient direction (IGO) features and specific energy functions. Then, the AAM model is updated accordingly. The Lukas Kanade algorithm is introduced in the iteration process. The test results based on the dataset of laser stripe images are shown in Fig. 5. It verifies that, based on the proposed PDM, the AAM model can realize feature point localization after iterations. This result can verify the effectiveness of the proposed laser stripe PDM. However, relying solely on this traditional method has lower stability under strong noise. Other methods need to be explored for more accurate and stable landmark positioning.

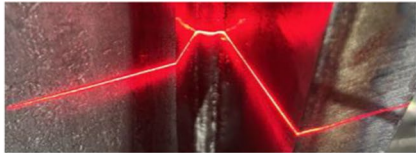
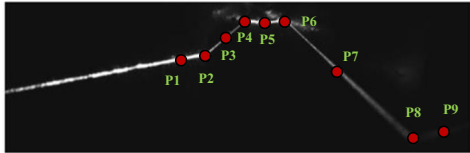
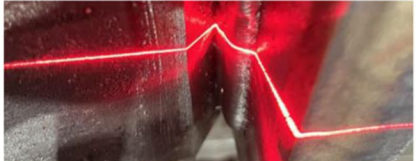
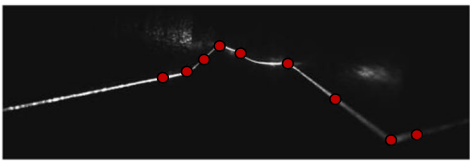
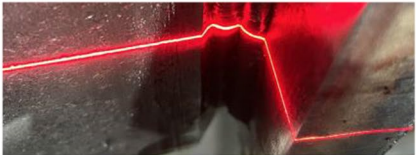
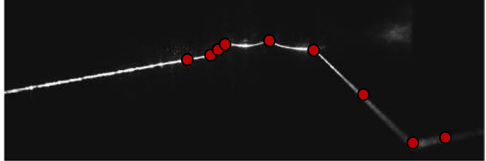
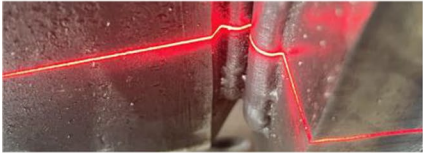
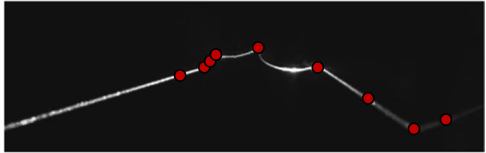
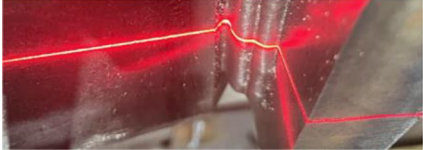
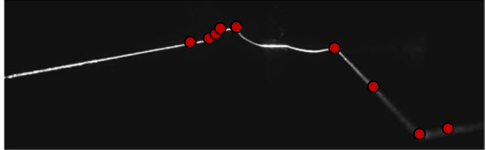
3.3 Feature point extraction algorithm

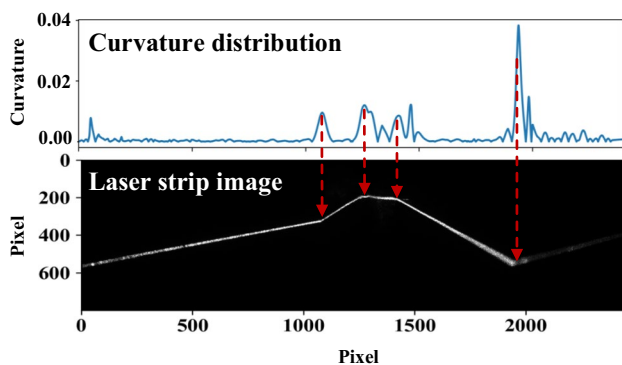
3.3.1 Overview of landmark positioning method

With the development of deep learning, the use of global and local high-dimensional convolutional features has significantly improved the performance of feature point localization. At present, there are two main regression methods for landmark positioning: heat map regression [19] and coordinate regression [20] [21]. The difference between the two methods lies in the different label types and the loss function forms, as shown in the flowchart of Fig. 6.

The heatmap regression-based coordinate positioning method requires converting the label coordinates corresponding to N images into the target of size $M * N$ in the form of a Gaussian heat map $h_1, h_2, \dots, h_N$. For the feature point $p_i = (x_i, y_i)$, a Gaussian distribution centered around it as $h_i(x, y) = \exp(-\frac{(x-x_i)^2 + (y-y_i)^2}{2\sigma^2})$ is generated. Therefore, taking HRNet [22] as an example, after inputting the training

Table 2 The demonstration of the feature point distribution of MLMP weld laser stripes

Welding layer and pass	Laser projected on the weld bead surface	Laser stripe image with feature points
Layer 2 Pass 1		
Layer 2 Pass 2		
Layer 3 Pass 1		
Layer 3 Pass 2		
Layer 3 Pass 3		

**Fig. 4** The extracted feature points based on second-order derivative

images and encoded labels into the network, the MSE error between heat maps is calculated as the training loss during the training process. The formula is

$$L_{MSE} = \sum_{i=1}^K \sum_{x=1}^M \sum_{y=1}^N \frac{[h_i(x, y) - \hat{h}_i(x, y)]^2}{K * M * N} \quad (2)$$

Among them, K is the number of feature points in a single image. M and N are the pixel sizes of Gaussian heat maps.

The coordinate regression-based method directly expands the high-dimensional image vector output by the network into a two-dimensional vector. Then an independent regression process on the coordinates of the K feature points of

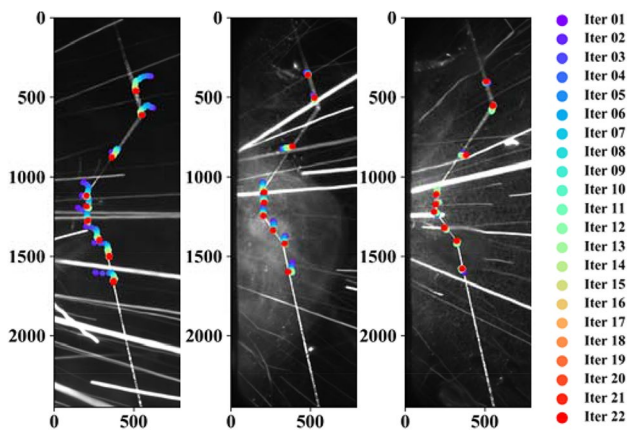


Fig. 5 Model verification based on iteration of the AAM model under different conditions

each image is performed. In the actual training process, for the i^{th} feature point $p_i = (x_i, y_i)$, the coordinate error is directly calculated and used as the training loss, and the formula is

$$L_{MSE} = \sum_{i=0}^K \frac{(x - x_i)^2 + (y - y_i)^2}{K} \quad (3)$$

Based on the preceding discussions, two types of methods have advantages and disadvantages. First, Gauss heatmap regression uses a gradual distribution around the target position to smooth the gradient and avoid the problem of over-fitting. Second, Gauss heatmap regression retains more correlation and local constraints between coordinate points, resulting in better convergence and generalization performance. However, in the process of heatmap decoding $\hat{m}, \hat{n} = \arg\max(\hat{h})$, the additional error is brought in due to the large difference in resolution. Lan et al. [23] proposed using a heatmap-in-heatmap (HIH) method to compensate for the quantization error of the down-sampling process by using additional heatmap coding coordinates.

But it trades higher computational effort for higher performance. Thus, the problem with the Gauss heatmap is the low efficiency and huge computational burden.

Research has been made to balance the efficiency and precision of the two types of methods. One popular route might be using a simplified coordinate error monitoring method to improve efficiency while keeping the global information. Xiong et al. [24] used a Band pooling module to convert the output Gauss heatmap into a Gauss vector in the vertical and horizontal sequences, which is used as a training loss-supervised metric. Jin et al. [25] proposed to remove the repeated up-sampling process in the traditional extraction method and directly achieve the coordinate regression based on the low-resolution feature heatmap.

Inspired by the above work, a laser stripe feature point tracking network (LSFPTN) is proposed in this paper (Fig. 7). The network uses the classical deep convolution network as the backbone for high-dimensional feature extraction. Also, to improve the robustness of laser stripe feature point extraction, the network optimizes the loss function of coordinate regression based on various strategies. The details will be discussed in the following subsections.

3.3.2 Network backbone

As a widely accepted deep neural network framework, the residual network (ResNet) [26] has achieved excellent results in different computer vision (CV) tasks, such as object detection and image classification. The introduction of the residual structure based on the skip-layer connections in deep networks brings multiple widely agreed advantages over others. The residual structure weakens the strong connection between layers, solving the problem of gradient disappearance and gradient explosion. Besides, the ResNet architecture allows a deeper network with fewer parameters, which avoids overfitting problems with high convergent

Fig. 6 Flow diagram of the heatmap regression model and the coordinate regression model

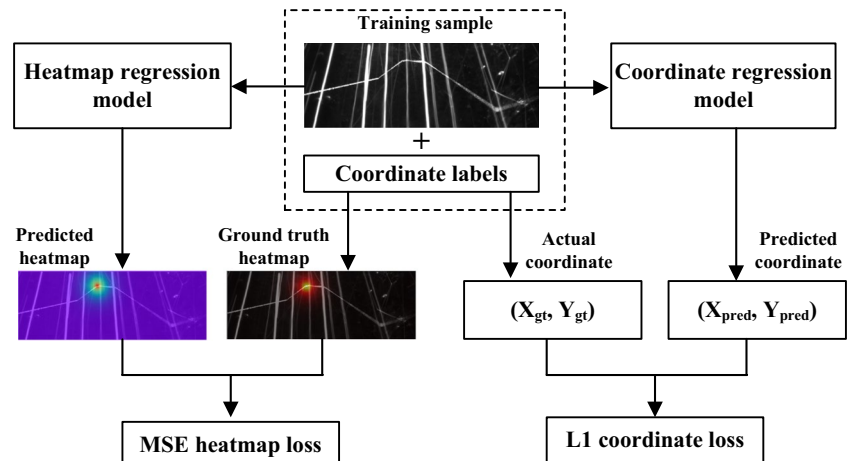
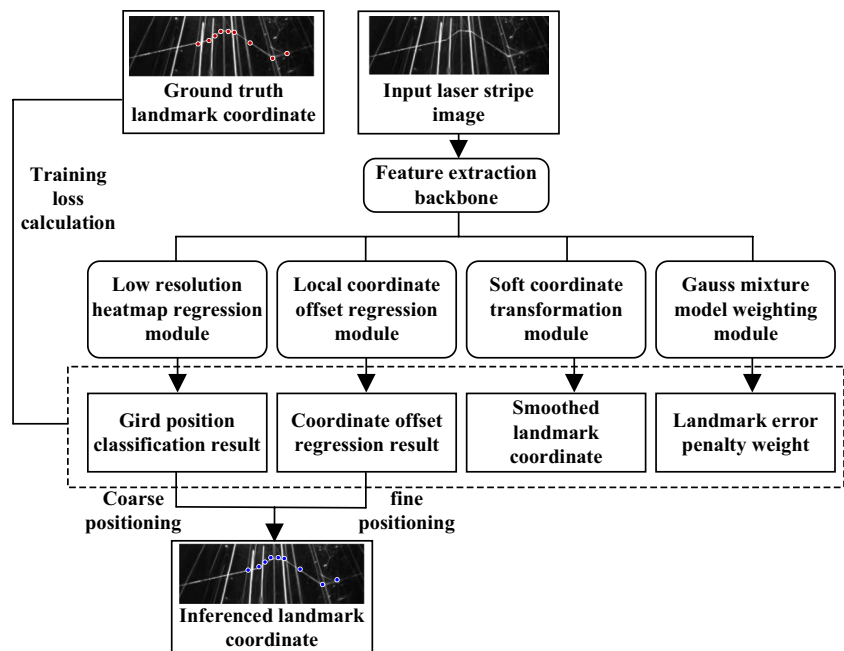


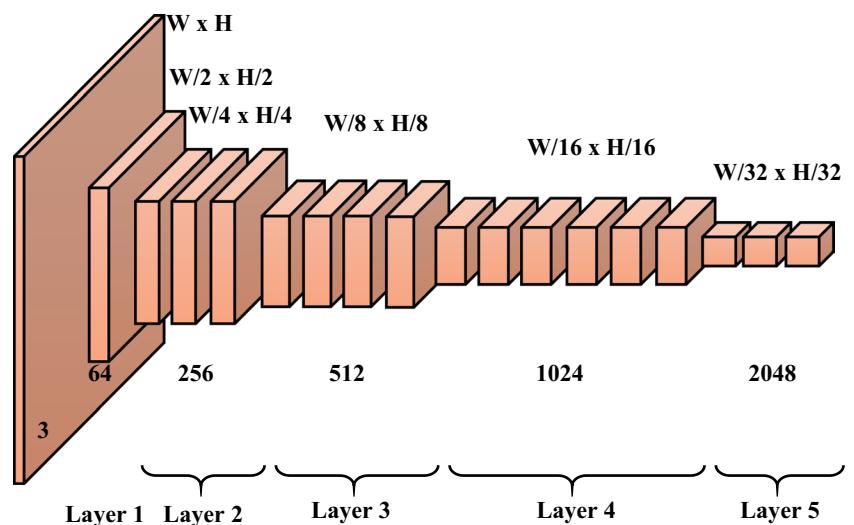
Fig. 7 The overall structure of the proposed method

speed and good generalization performance. Therefore, this paper uses ResNet as a feature extraction backbone network to do down-sampling operations based on the input images while preserving high-dimensional image features.

The overall structure of the backbone network is shown in Fig. 8. The whole model consists of five convolution layers. The output of higher dimensional features is obtained by different stages. Each convolutional layer mainly includes the stack of basic Bottleneck (BTN) modules. The structure of layer 1 consists of a regular convolution layer, batch normalization layer, and maximum pooling layer. Layers 2–5 have a similar structure, including several BTN modules, respectively. The dimension of the input image is $(W, H, 3)$, and the output dimension of the first convolution layer is

$(\frac{W}{4}, \frac{H}{4}, 64)$. To avoid the information loss of the convolution and pooling in the first layer, the loss-oriented learning and the further down sampling are required.

The squeeze-and-excitation network (Senet) [27] applies attention mechanisms to the dimension of feature channels and realizes feature dimension modeling. Through training, the weight of important features in the global feature channel can be increased, which can effectively improve network performance. Therefore, based on the two BTN structures of Resnet, Senet is stacked as a feature selector. The Senet module is placed before the superposition of residual and short connections to form a new SE-BTN structure, as shown in Fig. 9. Then the original BTN structures are replaced by the SE-BTN structure in each layer.

Fig. 8 The backbone of the proposed landmark detection network

3.3.3 Composition of loss function

After the global down-sampling feature extraction, sufficient landmark positioning information can be retained from the laser stripe image, so subsequent up-sampling structures need to be used to achieve the final coordinate regression. To supervise the optimization process of model parameter optimization, the loss function mainly consists of two parts: the basic loss function and the extended loss function. The basic loss function ensures that the model retains global and local constraint information while minimizing computational burden as much as possible. The extended loss function is weld feature points tracking oriented and stabilizes the overall model training process.

Low-resolution heat map loss function To preserve the high-precision positioning performance brought by the heat map regression and avoid the burden of excessive computation, a “coarse-to-fine” feature point positioning strategy is implemented. The model resizes the input images to $W_o \times H_o$, so it can adapt to different forms of laser images during the welding process. Then, the input laser image needs to be evenly divided into grid blocks of size $W * H (W = \frac{W_o}{stride}, H = \frac{H_o}{stride})$ according to a specific grid width $stride$.

In the first step of the coarse positioning process, based on the high-dimensional features, a convolutional up-sampling layer is used to obtain a Gaussian heatmap corresponding to the current feature point landmark. Then the grid coordinates with the highest heatmap value are obtained, which means the target feature point position is roughly controlled within a square area with the area of $stride \times stride$. As shown in Fig. 10, this process is equivalent to finding the corresponding grid positions in a total of $W * H$ grids, so it is essentially a multi-classification problem. Thus, the L2 loss function is used to calculate the mean square error (MSE) of the predicted results and target position. Although this strategy still retains the advantages of Gaussian heat maps for prediction, the computational complexity is reduced by two times in this coarse positioning process.

The landmark point $p_i(x_{ijh}, y_{ijh})$ of the current row j and column h , the value of the predicted heat map output by the network is $\hat{H}_i(x_{ijh}, y_{ijh})$, and the value of the ground truth label transcoded heat map is $H_i(x_{ijh}, y_{ijh})$. Then, based on the heat map distribution results, the coarse positioning error of K feature point landmarks L_{coarse} in the image can be calculated. This loss monitors the precision of coarse coordinate positioning in the image grid based on the generated heatmap during the training process, and its formula is as follows:

$$L_{coarse} = \sum_{i=1}^K \sum_{j=1}^W \sum_{h=1}^H \frac{(H_i(x_{ijh}, y_{ijh}) - \hat{H}_i(x_{ijh}, y_{ijh}))^2}{K * W * H}, H_i(x_{ijh}, y_{ijh}) \in \{0, 1\} \quad (4)$$

Wing loss-based local constraint loss function Based on the coarse positioning results, to achieve high accuracy in landmark feature point positioning, the second step “precise positioning” process needs to be carried out. The current feature point $p_i(x_i, y_i)$ has already been predicted in a certain grid. Therefore, on this basis, the problems of steep gradient changes and poor generalization ability can be solved by introducing the traditional coordinate regression method as a training method. In the grid, the relative error from the grid standard zero point to the current predicted position is calculated as the training supervision loss.

To solve the information loss problem of local constraints in coordinate regression, a local constraint module has been added to further improve the precision and stability of landmark coordinate prediction. Therefore, based on the defined PDM, the nearest M feature point coordinate will be calculated simultaneously when processing the precise positioning of the current landmark point $p_i(x_i, y_i)$. Since the distribution of laser stripe points is basically along the single-axis direction of the image, the Euclidean distance is used as a proximity measure. During the precise positioning

Fig. 9 The modified SE-BTN block in each layer of the backbone

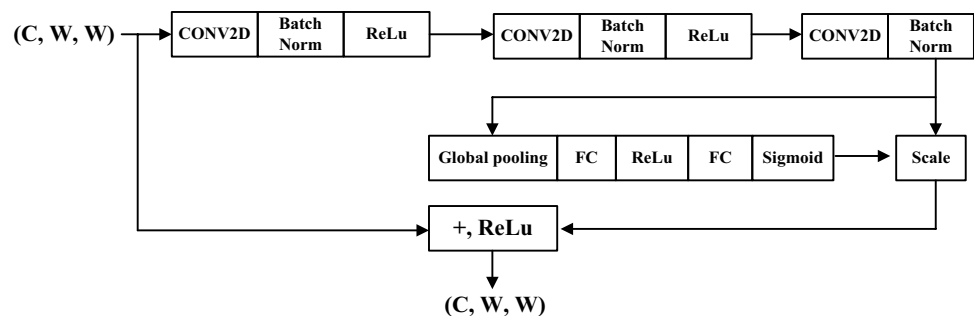


Fig. 10 Demonstration of the low-resolution heatmap training and predicting process

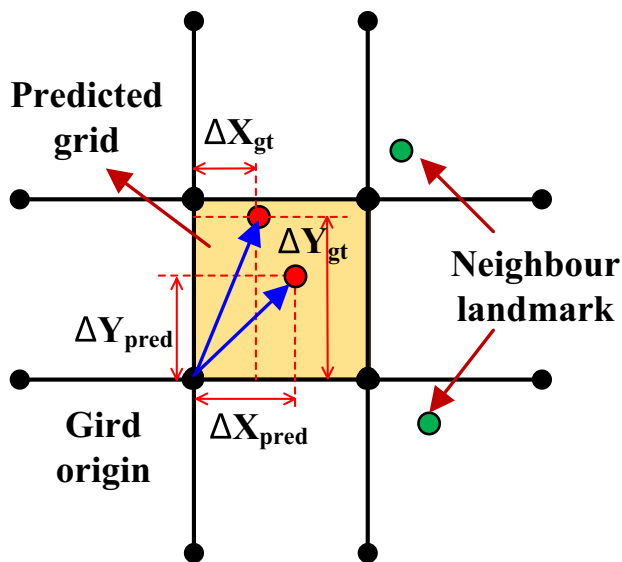
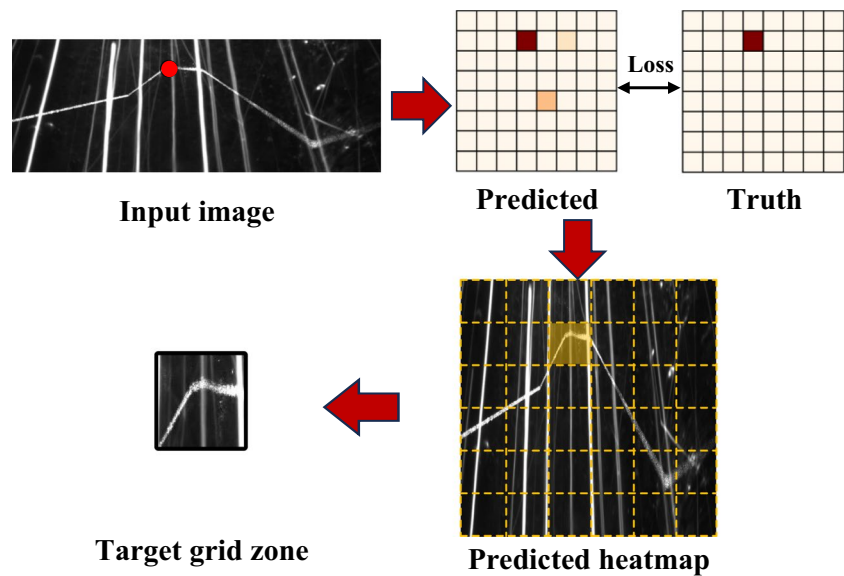


Fig. 11 The principle of the calculation of local constrain offset loss

process, as shown in Fig. 11, the errors of the offset values of the current landmark and adjacent landmark in the X and Y directions are both calculated and used as training supervision parameters.

The L1 loss is usually selected as the loss function of regression prediction. In the grid box, the coordinate point positioning error is relatively large in the early stage of training, and the regression results can be effectively optimized with gradient descent. However, during the middle and later stages of training, the overall coordinate regression L1 error is relatively small, resulting in a smaller training gradient. Thus, the training efficiency is seriously affected. To achieve

higher prediction precision, a more effective loss function needs to be implemented. Wing loss [28] is a variant of L1 loss. The principle is using a logarithmic function near the zero position; it leads to large gradients even for small loss. Therefore, it can solve the above problems in the later training stage. The formula is shown in the following:

$$\text{wing}(x) = \begin{cases} \omega \ln \left(1 + \frac{|x|}{\epsilon} \right), & \text{if } x < \omega \\ |x| - C, & \text{otherwise} \end{cases} \quad (5)$$

Among them, wing loss limits the nonlinear part within the range of $[-\omega, \omega]$. ϵ is used to control the curvature of nonlinear regions, $C = \omega - \omega \ln(1 + \frac{\omega}{\epsilon})$.

The offset value in the X and Y directions between the predicted results and the ground truth result L_{fine} is calculated through the following formula:

$$L_{fine} = \sum_{i=1}^K \sum_{H_i(x_i, y_i)=1} \left(|\text{wing}(ex_i)| + |\text{wing}(ey_i)| \right) / 2K \quad (6)$$

$$ex_i = Ox_i - Ox'_i, ey_i = Oy_i - Oy'_i \quad (7)$$

Among them, ex_i and ey_i are the errors of the relative coordinate offset values in the X and Y directions predicted by the network, Ox_i , Ox'_i , Oy_i , and Oy'_i are the actual and predicted coordinate offset values in the X and Y directions, respectively.

To further increase the local constraints of points in PDM, the positioning error of neighboring points is also calculated as a loss to supervise the training process. Based on the neighboring points regression, the connections between the near points are enhanced, therefore, to improve the tracking accuracy under strong noise conditions, the precise positioning offset error $L_{neighbor}$ of M neighboring points of the current landmark is given by the following formula:

$$L_{neighbor} = \sum_{i=1}^N \sum_{m=1}^M \sum_{H_i(x_i, y_i)=1} (|wing(nex_{im})| + |wing(ney_{im})|) / 2MN \quad (8)$$

$$nex_{im} = nOx_{im} - nOx'_{im}, ney_{im} = nOy_{im} - nOy'_{im} \quad (9)$$

Among them, nex_{im} and ney_{im} are the errors of the relative coordinate offset value of the m th neighboring landmark predicted by the network in the X and Y directions; nOx_{im} , nOx'_{im} , nOy_{im} , and nOy'_{im} are the actual and predicted coordinate offset values in the X and Y directions, respectively.

Soft coordinate loss function Although this network structure uses the grid heat map in the coarse positioning module, it essentially still uses the idea of coordinate regression in the precise positioning stage. During the training process, the initial predicted coordinate loss will decrease relatively fast in the gradient direction, but the loss in the middle and later stages is often difficult to continue to decrease. This is due to the poor performance of hard label training in coordinate regression. Fard et al. [29] proposed an auxiliary loss function that converts hard labels of real coordinates into smooth distributed soft labels based on the ASM. The main idea is to first learn the soft label distribution, which is easier to achieve, and then gradually increase the weight of the hard label as the training progresses. It has the potential to guide the training loss reduction in a way closer to the Gaussian heat map.

For a shape S consisting of n points,

$$S = \{(x_1, y_1), \dots, (x_n, y_n)\} \quad (10)$$

To establish simplified shape space components, PCA needs to be performed on a large number of shape samples to obtain the q eigenvectors $P = (p_1, p_2, \dots, p_q)$ of their covariance matrix. Simultaneously, the average S of shape samples is obtained. Based on this model, the current shape S can be represented:

$$S \approx \bar{S} + Pb \quad (11)$$

where b is the q -dimensional vector corresponding to the spatial feature vector P :

$$b = \{\varepsilon_1, \varepsilon_2, \dots, \varepsilon_q\} \quad (12)$$

According to the above formula, by generating a q -dimensional vector b containing different elements, a new set of smooth shape data can be randomly generated based on the original training shape samples. Thereby a deformable model can be established. However, after the transformation operation of this module, the differences between the generated new shape and the actual shape should be suppressed. Therefore, the variance va of the elements ε_i in vector b is used as the control parameter: $\varepsilon_i \in [-\sqrt{3}va, \sqrt{3}va]$. Based

on the above process, a new deformable shape model can be obtained:

$$S_{changed} = \bar{S} + P\tilde{b} \quad (13)$$

Therefore, the above transformation process is described as smooth coordinate transformation (SCT), which converts the input coordinate point set $S_{original}$ into a new point set S_{smooth} , as a soft label for model training. After performing the above transformation on the ground truth coordinates (Cx_i, Cy_i) of the i th landmark in the image, a smooth coordinate point (SCx_i, SCy_i) is obtained. Then, based on the predicted coordinate point coordinates output (Cx'_i, Cy'_i) in Section 4.3.2, the loss of the soft label based on SCT can be calculated according to the following formula, and the whole process is shown in Fig. 12:

$$L_{ASM} = \sum_{i=1}^K (|SCx_i - Cx'_i| + |SCy_i - Cy'_i|) / 2K \quad (14)$$

The soft supervision loss is added as an auxiliary loss to better learn the gradient descent direction in the beginning stage of training. However, in the later stage of training, the weight of the soft coordinate loss should be reduced to improve the precision of the actual coordinate position and avoid oscillations in the convergence process. As the training epoch increases, the weight gradually decreases.

Penalty emphasis on target tracking point Unlike other human pose estimation and facial recognition tasks, the goal of this model is not to accurately locate every landmark feature point, but to minimize the error of key feature points required for tracking. Therefore, in the above loss calculation process, different weight parameters need to be set for each feature point. As shown in Fig. 13, the weight parameters are set according to the Gaussian distribution centered on two central key feature points. Secondly, the local constraint mode should be considered, as the precision of neighborhood points will affect each other due to the neighborhood prediction module. Therefore, the two can be mixed based on a mixed Gaussian model, with the following formula:

$$P(y|\theta) = \sum_{d=1}^D \alpha_d \varnothing(y|\theta_d) \quad (15)$$

Among them, α_d is the weight of the d th Gaussian distribution, $\varnothing(y|\theta_d)$ represents the d th Gaussian distribution model, $\theta_d = (\mu_d, \sigma_d)$. The number of key feature points in this model: $D=2$, $\alpha_1 = \alpha_2 = 0.5$.

Therefore, based on the above loss functions, the overall loss function of this model is

$$L_{LSTPTN} = L_{coarse} + \alpha * L_{fine} + \beta * L_{neighbor} + L_{asm} \quad (16)$$

Then, α and β are the proportional coefficients of the offset error.

Fig. 12 Flow chart of the calculation process of soft coordinate loss

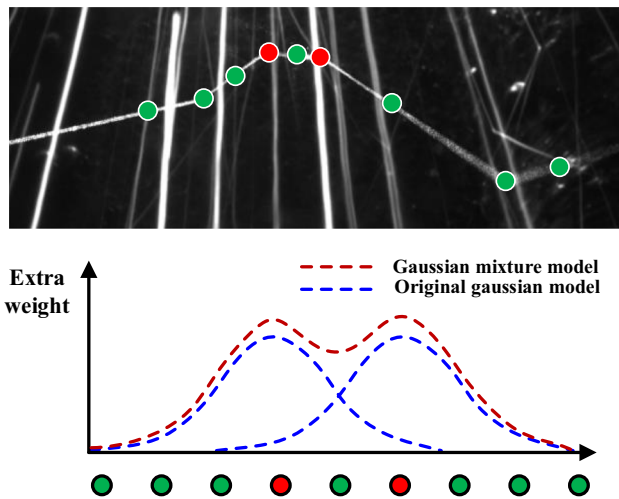
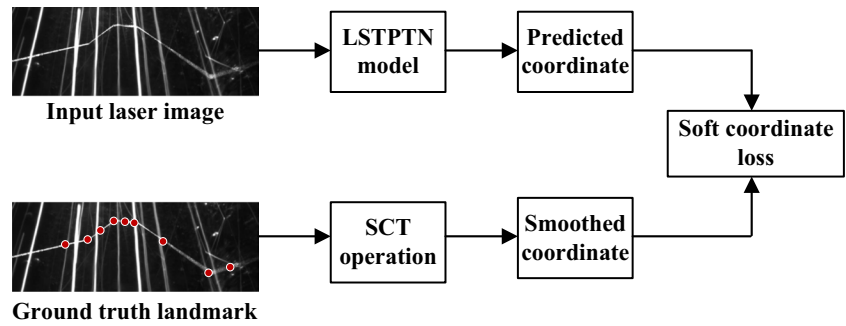


Fig. 13 Weight distribution of the feature landmark points

4 Extension of the proposed feature extraction method

4.1 Basic network parameter tuning

To achieve higher precision in seam feature point extraction, it is necessary to optimize the internal parameters of the LSFPTN model. The training process and internal structure parameters of the network have a significant impact on the performance of the model. In the experiment, only the basic loss function was used in these experiments. Explorations were conducted on the backbone network for feature extraction, the size of the training batch, the number of predicted neighboring points, and the weight ratio of the “coarse-to-fine” landmark positioning loss. The specific parameters and performance of the test are shown in Table 3.

During the experiment, some mainstream deep networks (ResNet50, ResNet18, and MobileV2) were tested as backbones for feature extraction, because there are similar efficient residual structures in its network structure (inverted residual connections for MobileNet). It was found that ResNet50 had relatively high accuracy in

feature point extraction, with an average absolute error of 3.24 pixels. On this basis, different batch numbers were selected as parameters for the experiment, and the batch size was set from 4 to 32 based on the platform computing capacity. From the results, the feature point extraction error first decreased and then increased. The highest model precision can be achieved when the batch size is 8. Also, the number of predicted adjacent feature points is set to 2, 4, and 6. When the number of prediction points is 4, it achieves the highest accuracy. It may be because too many prediction points lead to increased network burden, while fewer prediction points result in slightly weaker local constraint ability. Finally, an experiment was conducted to obtain an appropriate loss weight ratio. When the weight ratio is 8, the network has the strongest ability to extract feature points. The results are shown in Fig. 14.

4.2 Overall process for MLMP seam tracking

To better adapt the model to the MLMP seam tracking process, some additional operations are required to make the whole process more complete. As shown in Fig. 15, for the raw laser stripe image, the automatic region of interest (ROI) extraction is performed first, followed by the feature point extraction process in the LSFPTN model. Finally, the target points are filtered to achieve the final 3D point output to the robot control cabinet.

4.2.1 Automatic ROI extraction

To reduce the computational burden and eliminate irrelevant interference, it is necessary to perform ROI extraction to achieve more efficient feature point extraction. The overall extraction process is shown in Fig. 16. Before welding arc initiation, a raw laser image is obtained and input to the proposed LSFPTN model for feature point extraction $S_{fp} = \{(x_1, y_1), \dots, (x_n, y_n)\}$. The feature point localization results are with high confidence under no welding noises, which can be used to set the original ROI bounding box $B = \{(X_{ul}, Y_{ul}), (X_{ur}, Y_{ur}), (X_{bl}, Y_{bl}), (X_{br}, Y_{br})\}$. Then, the calculated bounding box B_n of the extract feature points

is expanded outward evenly until reaching the designed target ROI size or approaching the original image boundary. Subsequently, during welding, the above process is repeated based on the feature point results from the previous frame. In addition, to avoid ROI drifting, a threshold for inter-frame movement of the bounding box is set Δm_{th} . The movement is represented by the difference in the center coordinate of B , which is $P_{center} = ((X_{ul} + X_{br})/2, (Y_{ul} + Y_{br})/2)$. When the movement exceeds the threshold ($\Delta m = P_{center}^{n+1} - P_{center}^n \geq \Delta m_{th}$), the ROI area of the previous frame is directly used for subsequent extraction, since the movement of target laser stripes between adjacent frames is relatively small. Based on this procedure, the automatic ROI extraction is achieved with no extra computation.

4.2.2 Filtering of output feature point

As mentioned, the neighborhood coordinate regression module was added to strengthen the local constraints of the PDM. Therefore, after inputting ROI laser images into the LSFTN model, two different detection results can be obtained. The first is the direct positioning result for the current target point (P4 for layer 2 pass 2). The other is the fusion of localization results M_2 from other neighboring point coordinate regression. Although the difference in coordinates is relatively small in testing, to ensure the accuracy of the output points, the 2D Kalman filtering was used to further optimize the target point coordinates.

Table 3 The process of the network parameter tuning

Index	Net structure				Precision/pixel		
	Backbone	Batch size	Neighbor number	Weight ratio	RMSE	MAE	ME
1	R50	8	4	8	3.61	3.24	5.51
2	R50	8	4	10	3.89	3.65	5.70
3	R50	16	4	8	3.83	3.60	5.67
4	R50	4	4	8	3.87	3.64	5.63
5	R50	32	4	8	3.90	3.65	5.61
6	R50	8	6	8	3.74	3.43	5.69
7	R50	8	2	8	3.81	3.55	5.67
8	Mv2	8	4	8	3.74	3.42	5.70
9	R18	8	4	8	3.82	3.58	5.64
10	R50	8	4	6	3.73	3.47	5.67

Fig. 14 The test results of the proposed LSFTN model with adjusting model parameters: **a** backbone, **b** batch size, **c** neighbor number, and **d** weight ratio

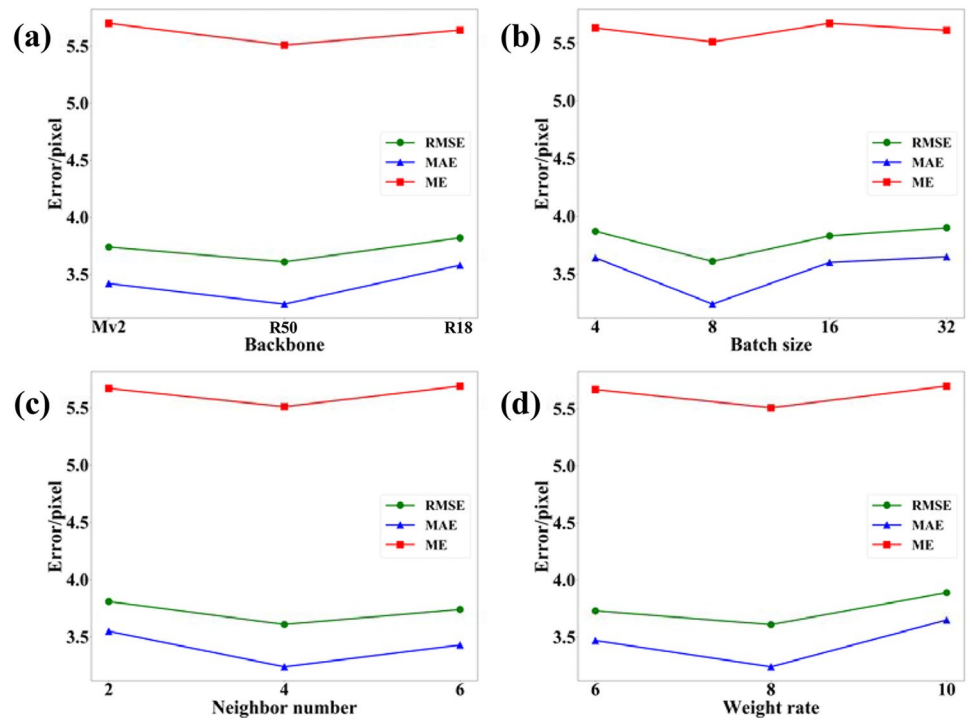
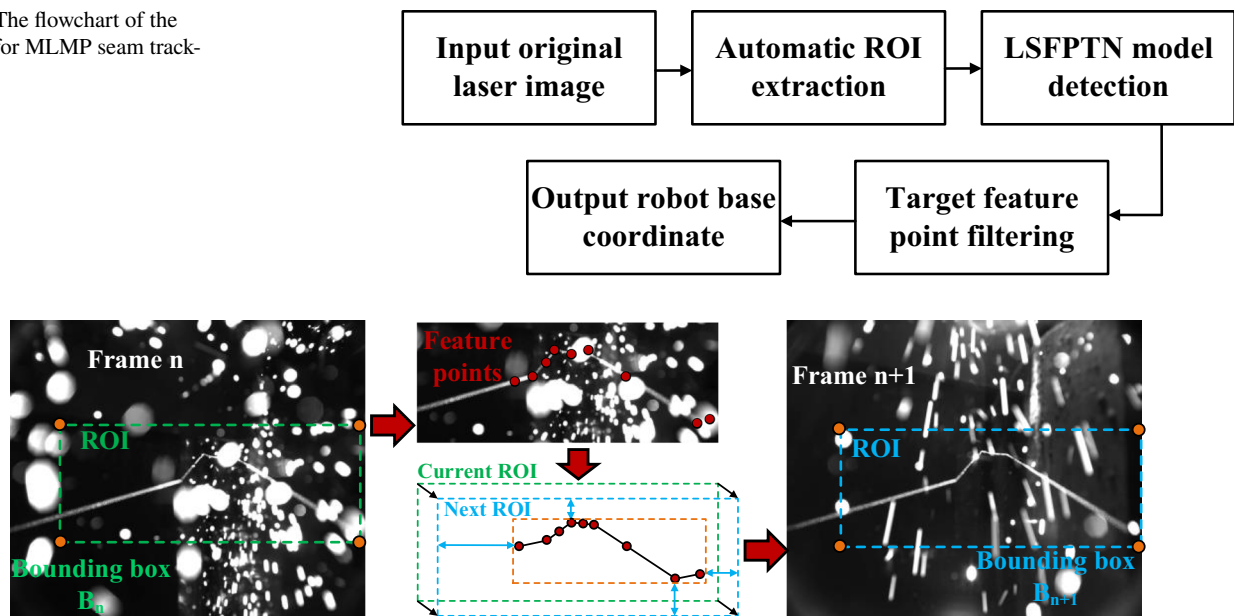


Fig. 15 The flowchart of the process for MLMP seam tracking**Fig. 16** The flowchart of ROI extraction

During the welding process, the two t th target point coordinate states are marked as $C_t^1 = (X_t^1, Y_t^1, dX_t^1, dY_t^1)^T$ and $C_t^2 = (X_t^2, Y_t^2, dX_t^2, dY_t^2)^T$, with the covariance matrix P^1 and P^2 . The positioning noise covariance matrix is set as R^1 and R^2 . The process noise covariance matrices are Q^1 and Q^2 . The different positioning results are M_1 and M_2 . Taking the C_t^1 as an example, the prediction is $C_t^1 \hat{t} = A * C_t^1$. The covariance matrix $P^1 \hat{t} = A * P^1 * A^T + Q^1$. Then, in the updating process, the Kalman gain is obtained $K^1 = P^1 \hat{t} / (P^1 \hat{t} + R^1)$. The target point state can be updated: $C_{t+1}^1 = C_t^1 \hat{t} + K^1 * ([M_1, M_2] - H * C_t^1 \hat{t})$ and $P^1 = (I - K^1 * H) * P^1 \hat{t}$. Similarly, the C_{t+1}^2 and P^2 are also obtained.

To get the final optimal target point coordinate state C_{t+1} , the inverse of the measurement noise covariance is used as a weight, since the lower noise leads to higher confidence. Thus, $C_{t+1} = (\omega_1 * C_{t+1}^1 + \omega_2 * C_{t+1}^2) / (\omega_1 + \omega_2)$, where the $\omega_1 = 1/R^1$ and $\omega_2 = 1/R^2$. Based on this procedure, the final state can be regarded as the best estimate of the point coordinates based on both positioning results.

5 Experimental result and discussion

5.1 Experimental scheme

Firstly, to verify the effectiveness of the proposed LSFPTN model, this paper tests the laser stripe feature point positioning performance based on the PDM. Also, ablation experiments were designed to evaluate the rationality and feasibility of each module of the model and the modification strategies.

Table 4 The platform setup for model training and testing

GPU	CPU	Memory
RTX 2060	AMD 6-core	6 GB

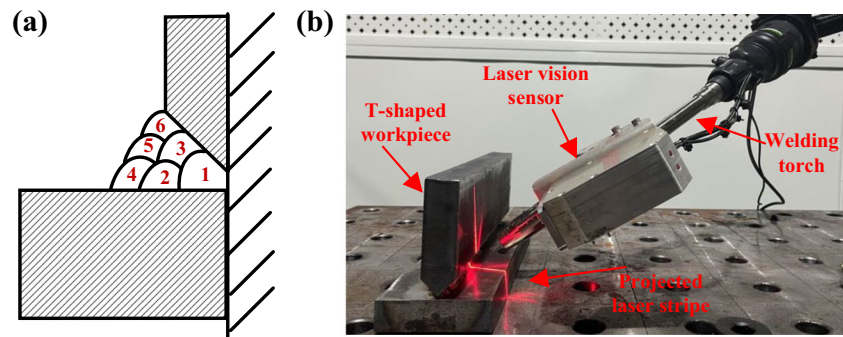
Table 5 The details of the welding plate

Welding parameters	Value
Plate material	Q345 steel
Plate geometry	300 * 100 * 30 mm
Groove type	45° K-shaped groove
Welding current (A)	140–180
Welding voltage (V)	22–26
Welding speed (cm/min)	15–20
Gas flow (L/min)	18
Welding mode	DC

Finally, comparative experiments were conducted with other feature point prediction algorithms to further validate the effectiveness. The training and testing platform settings for the network are shown in Table 4.

Secondly, to further validate the proposed MLMP weld seam feature point tracking method, experiments were conducted on the robotic welding system introduced in Section 2. The welding materials and welding parameters are shown in Table 5. The planning strategy for MLMP welding is shown in Fig. 17 (a), consisting of three layers and six passes, including one root pass and five filling passes.

Fig. 17 The MLMP welding experiment: **a** the MLMP planning strategy and **b** the experimental setup



5.2 Accuracy verification of the proposed method

To verify the accuracy of the proposed tracking method, an MLMP welding experiment was conducted according to the above experimental scheme. Some typical laser stripe images with the detected feature points from each welding pass are listed in Fig. 18 (a). It has been verified that the tracking points can be successfully extracted. Figure 18 (b) shows the macroscopic images of the weld joint in the MLMP welding process. According to the tracking error for

each pass in Table 6, the conclusion can be drawn that the proposed method can realize accurate MLMP seam tracking. Layer 2 pass 1 presents the minimum MAE of 0.187 mm, and the maximum MAE occurs in the last pass of layer 3. The tracking error shows a rising trend as the number of welding layers and passes increases. This is probably because the ambiguity of the weld surface shape increases when the MLMP welding progresses to the middle and later stages. However, the results still support that the precision can meet the requirement of MLMP seam tracking.

Fig. 18 The result of the robotic MLMP welding experiment: **a** the tracking performance and **b** the macroscopic image of each welding pass

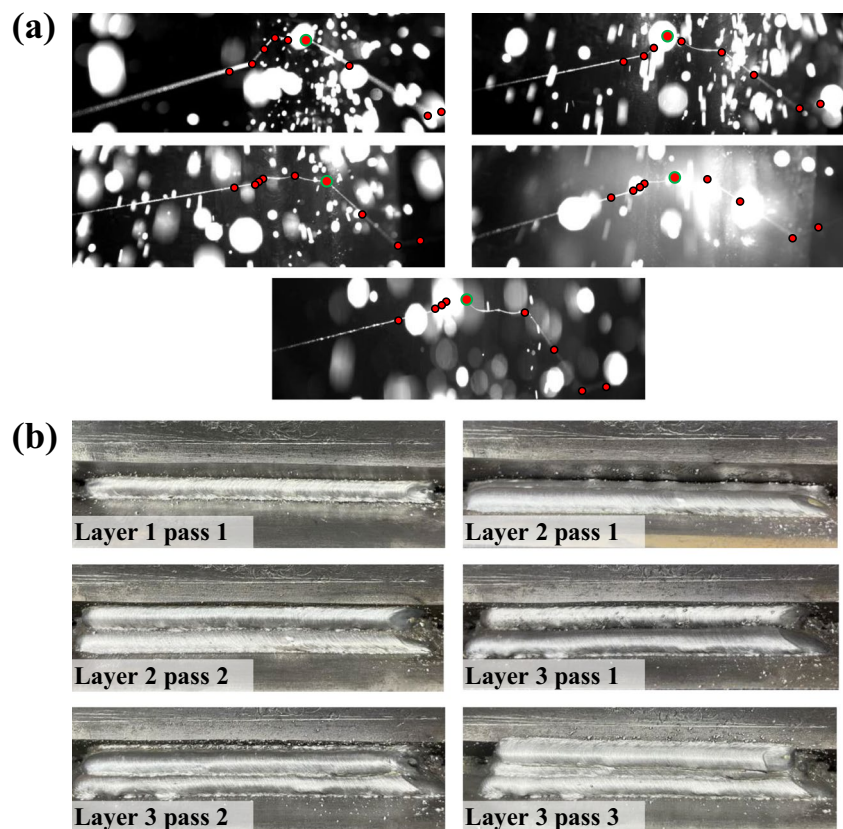


Table 6 The tracking result based on the proposed method for each pass

Welding layer and pass	Precision/mm		
	RMSE	MAE	ME
Layer 2 pass 1	0.260	0.187	0.593
Layer 2 pass 2	0.276	0.214	0.591
Layer 3 pass 1	0.258	0.190	0.620
Layer 3 pass 2	0.283	0.266	0.648
Layer 3 pass 3	0.313	0.287	0.656

Table 7 Results of comparison experiment with other feature backbone models

Model backbone	Precision/pixel		
	RMSE	MAE	ME
VGGNet	3.97	3.62	6.09
Swin transformer	3.46	3.28	5.12
R0	3.67	3.47	5.77
Proposed	2.35	1.90	4.59

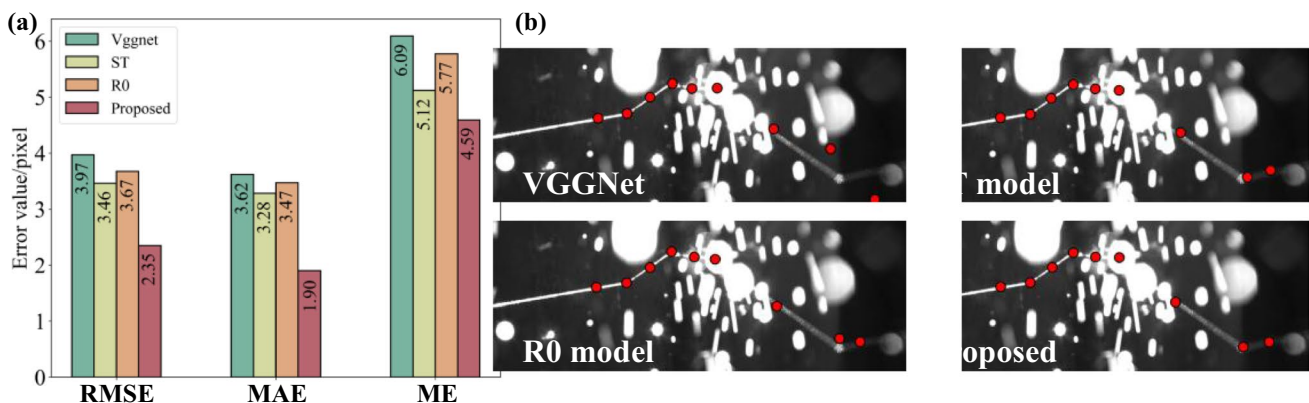
5.3 Effectiveness verification of the proposed model

Verification of proposed model backbone To verify the feature extraction ability of the backbone in the proposed LSFPN model, comparative experiments were conducted with other mainstream popular feature extraction models. VGGNet [30], as a classic convolutional neural network, is widely used in various CV tasks due to the characteristics of shared weights and excellent generalization performance. Currently, transformer-based models have achieved global feature modeling of images, providing a new paradigm for the CV field. Therefore, the state-of-the-art Swin

transformer (ST) [31] was introduced into the comparison experiment. In addition, the original Resnet network without SE-BTN was also used as the baseline (marked as the R0 model). The experimental results are shown in Table 7.

According to the experimental results in Fig. 19 (a), the proposed model exhibited the lowest level of positioning error. The positioning error of the VGGnet model feature is the highest, with an RMSE of 3.97 pixels. The performance in Fig. 19 (b) verified its susceptibility to welding interference. This may be due to the use of a large number of pooling layers in its structure, which reduces the spatial resolution of the network and leads to the loss of important location information. The features obtained by the R0 model achieved higher positioning performance (RMSE of 3.67 pixels), which proves that the addition of the residual connection strategy has a positive effect on improving precision. However, the proposed model backbone achieved a higher accuracy with an error decrease of 35.97%, verifying the effectiveness of the proposed SE-BTN module in the model feature extraction. The ST model based on the transformer achieved better performance than the R0 model, but its error level is still higher than the proposed model. In Fig. 19 (b), the proposed model exhibits better stability and robustness under high noise conditions. This may be due to the more complex and highly nonlinear structure of the ST. Although it brings strong fitting ability, it is more suitable for conditions with large-scale data with rich information. The former mixes local and global information at each level, while the latter strictly extracts global information from local features, which may be more suitable for the proposed “coarse-to-fine” laser feature point positioning process.

Verification of proposed “coarse-to-fine” strategy To verify the proposed “coarse-to-fine” strategy, a series of experiments were conducted to verify the feature point positioning ability of the proposed LSFPN model, where the wing loss was used to replace the traditional L1 loss, and soft

**Fig. 19** The performance comparison based on different extraction backbones: the (a) pixel error result and (b) feature point positioning result

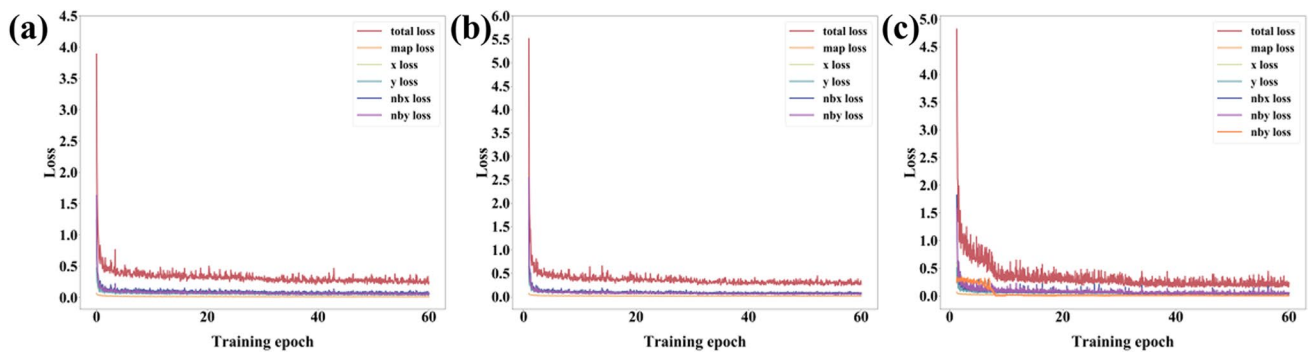


Fig. 20 The comparison of training loss with different loss function compositions **a** with traditional L1 loss, **b** with wing loss, and **c** with ASM loss

coordinate loss and feature point weighted loss based on the Gaussian mixture model were added to further optimize the training strategy. Therefore, ablation experiments for each part were conducted and the training loss composition curves of different models are shown in Fig. 20.

From the loss curve, based on the original L1 loss function, the fluctuation of various loss components during the training process is significant. As shown in Fig. 20 (b), with

the wing loss, the fluctuation amplitude of the overall loss decreases, indicating that the gradient descent direction during the training process is more stable. Therefore, it is proven that the loss calculation method can more effectively supervise network parameter optimization. After adding the soft coordinate loss based on ASM, it shows a similar decline trend with other loss functions in the early training stage. In the middle stage, the loss curve slope of the original

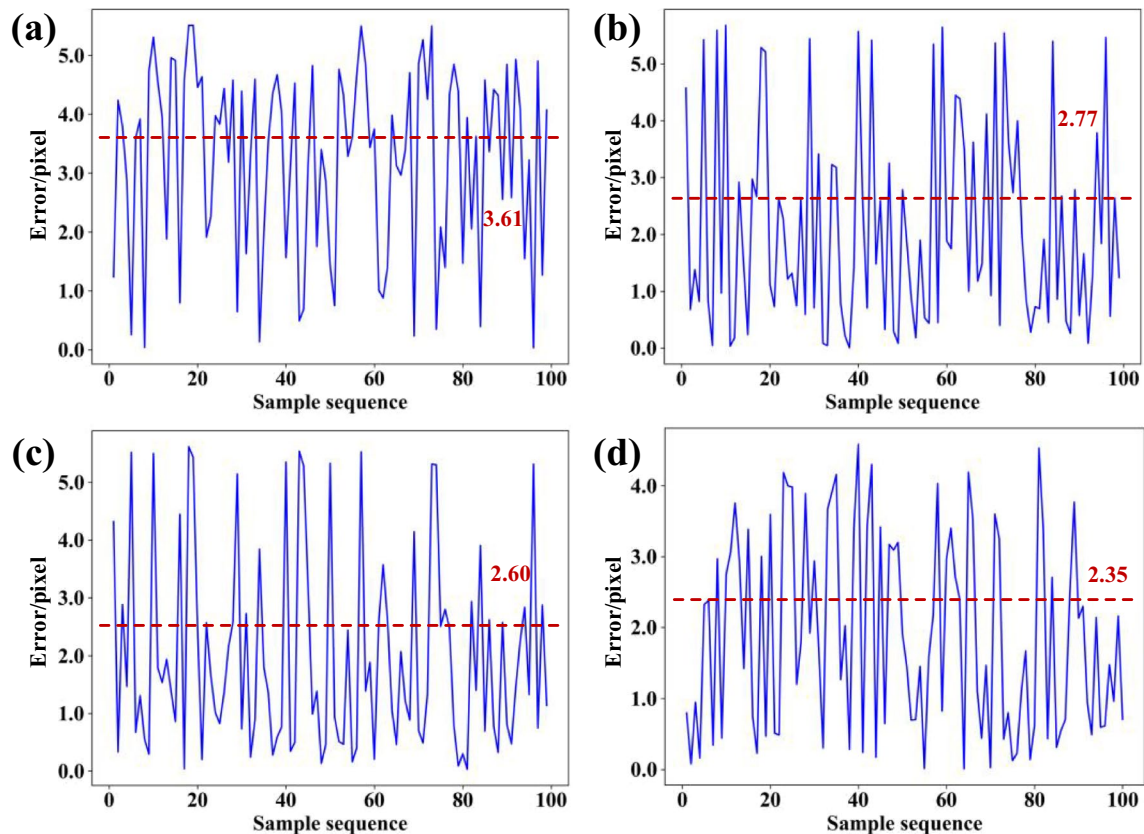


Fig. 21 The test result of ablation experiments with different loss function compositions **a** with traditional L1 loss, **b** with wing loss, **c** with weighted loss, and **d** with ASM loss

Fig. 22 The results of the proposed tracking method under strong environmental noise

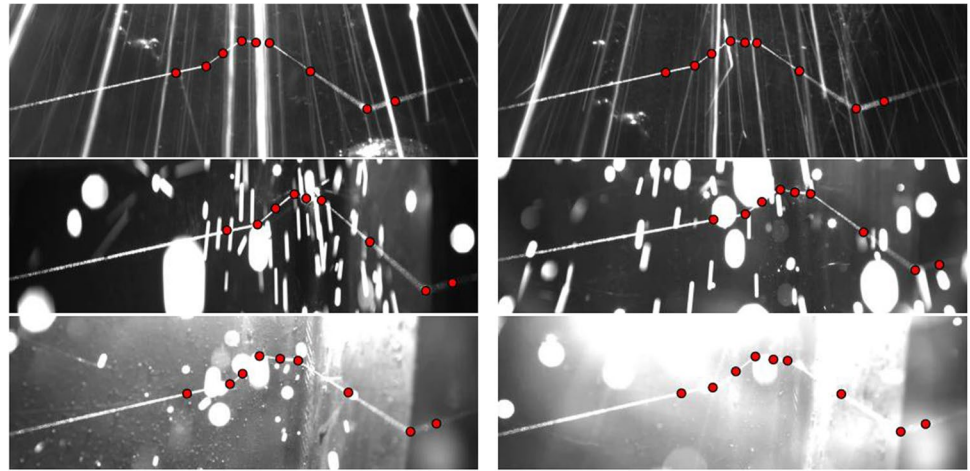


Table 8 Test result of the comparison experiments over other models

Feature extraction model	Precision/pixel		
	RMSE	MAE	ME
FPS3000	3.76	3.30	5.69
LSFPTN without extended loss	3.60	3.24	5.51
Proposed LSFPTN	2.35	1.90	4.59

model gradually decreases, while that of the modified model still decreases at a constant slope. Moreover, in comparison, the final overall loss presented a relatively lower level (Fig. 20 (c)). Therefore, it is verified that this loss function calculation method has the potential to make the training process stable and achieve lower error levels.

Figure 21 shows the prediction results of the trained model on the test dataset. The traditional L1 loss supervised network can achieve an RMSE of 3.61 pixels. And model trained with wing loss and feature point weighted loss controlled the RMSE within 3 pixels. Based on this, the addition of the soft coordinate loss can reduce RMSE to 2.35 pixels, which means the proposed model can still achieve precise feature point localization performance even under

strong noise conditions. Therefore, by using the “coarse-to-fine” landmark positioning strategy, this method preserves the advantages of heat map regression and realizes the unity of global convolution and local constraints. The experiment verifies that the optimized training method of this network can achieve higher accuracy in seam feature point extraction. Figure 22 presents part of the feature point extraction results of layer 2 and pass 1 based on the LSFPTN model.

5.4 Performance verification over other models

The FPS3000 [32] algorithm is a feature landmark positioning algorithm based on random forest (RF). An RF was set up for training each landmark point. Pairs of pixels are selected around the L landmarks and their pixel differences are used as the shape-indexed feature, which combined with the average shape $\bar{S}$ is the model input. In the t -level training process of RF, the split node parameters ϕ_l^t , $l = 1, 2, \dots, L$ are optimized. The sparse vector of the local binary feature (LBF) is the output. Then, based on a global linear regression trainer W^t , achieving the mapping from LBF to the predicted shape $\hat{S}$. Due to its high efficiency and accuracy, this algorithm has been widely applied in various face

Fig. 23 The results of comparative experiments: **a** the error curve of the FPS3000 algorithm and **b** the detection performance

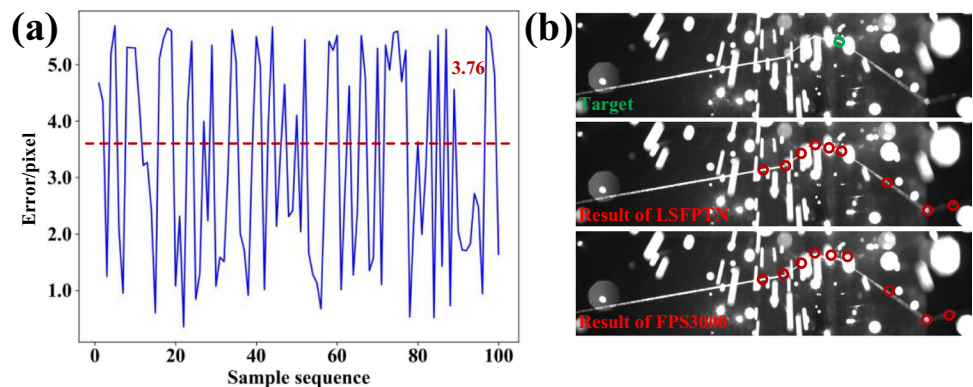


Table 9 Test result of the comparison experiments over other tracking methods

Tracking method	Precision/mm			Speed/FPS
	RMSE	MAE	ME	
DCFNet-based method	0.361	0.271	0.970	60
ECO-based method	0.339	0.273	0.743	15
KCF-based method	0.377	0.317	0.802	40
LSFPTN-based method	0.260	0.187	0.593	25

recognition tasks in recent years. This method was used to compare with the proposed method. The results are shown in Table 8.

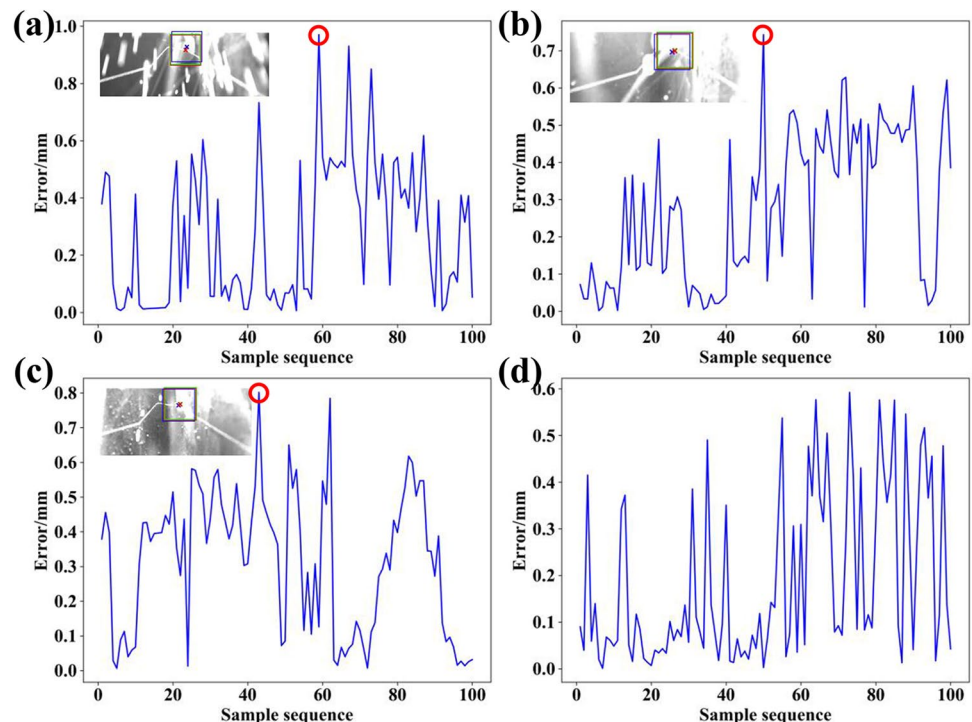
Then, the FPS3000 algorithm was compared with the proposed method on the same platform. From the experimental results, the feature point localization error of basic LSFPTN is smaller than the FPS3000 algorithm. This proves that the regression scheme of the basic network shows advantages over other regression methods. In addition, the proposed LSFPTN exhibits higher feature point extraction precision than the other three under different metrics. Comparing Figs. 23 and 21 (d), the RMSE of the former is 3.76 pixels, while that of the latter is approximately 2.35 pixels. The maximum positioning error has also decreased by about 1 pixel. The calculation speed of the proposed algorithm is approximately 25 FPS, although it is slower than the

comparative algorithm, it is more than sufficient to meet the requirements of real-time MLMP seam tracking. Therefore, the experiment verified that the proposed LSFPTN model exhibited higher performance than other models in laser stripe feature point extraction.

5.5 Performance verification over other tracking methods

To verify the tracking performance of the proposed method, comparative experiments were conducted based on traditional tracking methods, namely, the efficient convolution operation (ECO), the DCFNet, and the kernel correlation filter (KCF) method. The 3D tracking error results are shown in Table 9. The proposed method based on the LSFPTN model shows the highest precision with an RMSE of 0.260 mm. According to Fig. 24 (a)–(d), the other methods show relatively good performance in the beginning stage of welding. However, the stability and tracking precision were reduced as the welding noise got severe, especially when the welding splashed or arc light covered the target point of the laser stripe (Fig. 24 (e)). The experimental results indicate that persistent strong noise interference makes the other methods prone to tracking drift, leading to significant tracking errors in stages, while the proposed method has good robustness. It shows better advantages in both precision and speed in MLMP weld seam tracking.

Fig. 24 Tracking performance based on different methods: **a** DCFNet-based method, **b** ECO-based method, **c** KCF-based method, and **d** LSFPTN-based method



6 Conclusion

To solve the problems of feature point positioning in robotic MLMP weld seam tracking, this paper proposes an automatic feature point extraction method based on the feature PDM. The main contribution of this paper can be summarized as follows:

- (1) This article proposes a feature PDM for MLMP welds based on the shape characteristics of the laser stripes. The model describes the correlation between the shapes of laser stripes. The effectiveness of this model has been verified through experiments, laying the foundation for further MLMP seam tracking.
- (2) An automatic end-to-end laser stripe feature point tracking network model was designed. The model implements the feature extraction backbone with ResNet and attention mechanism. The “coarse-to-fine” positioning strategy with a low-resolution heat map and offset coordinate regression modules ensures global correlation and local constraints. Also, soft coordinate loss and Gaussian mixture model were added as extended loss functions to optimize the model performance. Based on the LSFPTN mode, a complete tracking process for MLMP seam tracking has been developed to ensure robustness, with the automatic ROI extraction method and the output point filtering.
- (3) The proposed method was verified with good precision and stability. Laser stripe feature points can be extracted accurately even under strong noises. The MAE of the 3D tracking error was controlled within 0.3 mm for each layer and pass. Also, the ablation experiments proved that the loss function modification can optimize network convergence. In comparative experiments, the proposed LSFPTN model exhibited better precision in the laser stripe feature extraction process. Compared with other tracking methods, the proposed tracking method showed higher noise resistance and smaller positioning errors.

Author contribution Conceptualization, methodology, validation, and writing—original draft, Fengjing Xu. Data curation, investigation, and resources, Lei He. Methodology, supervision, software, and resources, Zhen Hou and Runquan Xiao. Software, validation, and formal analysis, Tianyi Zuo and Jiacheng Li. Supervision, writing—review and editing, and resources, Yanling Xu. Conceptualization, writing—review and editing, and funding acquisition, Huajun Zhang. All authors have read and approved the final manuscript.

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Data availability The data sets are not exposed.

Declarations

Ethical approval The authors hereby state that the present work is in compliance with the ethical standards.

Consent for participation The authors declare that they all consent to participate in this research.

Consent for publication The authors declare that they all consent to publish the manuscript.

Competing interests The authors declare no competing interests.

References

1. Xu FJ, Xiao RQ, Hou Z, Xu YL, Zhang HJ, Chen, SB (2021) Multi-layer multi-pass welding of medium thickness plate: technologies, advances and future prospects. *transactions on intelligent welding manufacturing: Volume III No. 4* 2019, 3–33.
2. Xu FJ, Xu YL, Zhang HJ, Chen SB (2022) Application of sensing technology in intelligent robotic arc welding: a review. *J Manuf Process* 79:854–880
3. Wang NF, Zhong KF, Shi XD, Zhang XM (2020) A robust weld seam recognition method under heavy noise based on structured-light vision. *Robotics and Computer-Integrated Manufacturing* 61:101821
4. Zou YB, Chen XZ, Gong GJ, Li JC (2018) A seam tracking system based on a laser vision sensor. *Measurement* 127:489–500
5. Fan JF, Deng S, Ma YK, Zhou C, Jing FS, Tan M (2020) Seam feature point acquisition based on efficient convolution operator and particle filter in GMAW. *IEEE Trans Industr Inf* 17(2):1220–1230
6. Zou YB, Wei XZ, Chen JX (2020) Conditional generative adversarial network-based training image inpainting for laser vision seam tracking. *Opt Lasers Eng* 134:106140
7. Wu KY, Wang TQ, He JJ, Liu Y, Jia ZW (2020) Autonomous seam recognition and feature extraction for multi-pass welding based on laser stripe edge guidance network. *The International Journal of Advanced Manufacturing Technology* 111(9–10):2719–2731
8. He YS, Xu YL, Chen YX, Chen HB, Chen SB (2016) Weld seam profile detection and feature point extraction for multi-pass route planning based on visual attention model. *Robotics and Computer-Integrated Manufacturing* 37:251–261
9. Chen SF, Liu J, Chen B, Suo XY (2022) Universal fillet weld joint recognition and positioning for robot welding using structured light. *Robotics and Computer-Integrated Manufacturing* 74:102279
10. Xiao RQ, Xu YL, Hou Z, Chen C, Chen SB (2021) A feature extraction algorithm based on improved Snake model for multi-pass seam tracking in robotic arc welding. *J Manuf Process* 72:48–60
11. He YS, Ma GH, Chen SB (2021) Autonomous decision-making of welding position during multipass GMAW with T-joints: a Bayesian network approach. *IEEE Trans Industr Electron* 69(4):3909–3917
12. Zhao Z, Luo J, Wang YY, Bai LF, Han J (2021) Additive seam tracking technology based on laser vision. *The International Journal of Advanced Manufacturing Technology* 116(1–2):197–211
13. Yang GW, Wang YZ, Zhou N (2021) Detection of weld groove edge based on multilayer convolution neural network. *Measurement* 186:110129
14. Yang HL, Lyu JY, Cheng PJ, Tang XY (2021) Lddmm-face: large deformation diffeomorphic metric learning for flexible and consistent face alignment. *arXiv preprint arXiv:2108.00690*.

15. Li JF, Bian SY, Zeng AL, Wang C, Pang B, Liu, WT, Lu CW (2021) Human pose regression with residual log-likelihood estimation. In Proceedings of the IEEE/CVF international conference on computer vision (pp. 11025–11034).
16. Saragih JM, Lucey S, Cohn JF (2011) Deformable model fitting by regularized landmark mean-shift. *Int J Comput Vision* 91:200–215
17. Xiao RQ, Xu YL, Hou Z, Chen C, Chen SB (2022) An automatic calibration algorithm for laser vision sensor in robotic autonomous welding system. *Journal of Intelligent Manufacturing*, 1–14.
18. Milborrow S, Nicolls F (2008) Locating facial features with an extended active shape model. In *Computer Vision—ECCV 2008: 10th European Conference on Computer Vision, Marseille, France, October 12–18, 2008, Proceedings, Part IV 10* (pp. 504–513). Springer Berlin Heidelberg.
19. Geng ZG, Sun K, Xiao B, Zhang ZX, Wang JD (2021) Bottom-up human pose estimation via disentangled keypoint regression. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 14676–14686).
20. Fard AP, Mahoor MH (2022) Facial landmark points detection using knowledge distillation-based neural networks. *Comput Vis Image Underst* 215:103316
21. Xu ZX, Li BH, Yuan Y, Geng M (2021, May) Anchorface: an anchor-based facial landmark detector across large poses. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 35, No. 4, pp. 3092–3100).
22. Wang JD, Sun K, Cheng TH, Jiang BR, Deng CR, Zhao Y, Liu D, Mu YD, Tan MK, Wang XG, Liu WY, Xiao B (2020) Deep high-resolution representation learning for visual recognition. *IEEE Trans Pattern Anal Mach Intell* 43(10):3349–3364
23. Lan X, Hu QH, Chen Q, Xue J, Cheng J (2021) Hih: towards more accurate face alignment via heatmap in heatmap. *arXiv preprint arXiv:2104.03100*
24. Xiong YL, Zhou ZJ, Dou YH, Su ZZ (2020) Gaussian vector: an efficient solution for facial landmark detection. In Proceedings of the Asian Conference on Computer Vision
25. Jin HB, Liao SC, Shao L (2021) Pixel-in-pixel net: towards efficient facial landmark detection in the wild. *Int J Comput Vision* 129:3174–3194
26. He KM, Zhang XY, Ren SQ, Sun J (2016) Deep residual learning for image recognition. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 770–778)
27. Jin X, Xie YP, Wei XS, Zhao BR, Chen ZM, Tan XY (2022) Delving deep into spatial pooling for squeeze-and-excitation networks. *Pattern Recogn* 121:108159
28. Feng ZH., Kittler J, Awais M, Huber P, Wu XJ (2018) Wing loss for robust facial landmark localisation with convolutional neural networks. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 2235–2245)
29. Fard AP, Abdollahi H, Mahoor M (2021) ASMNet: a lightweight deep neural network for face alignment and pose estimation. In Proceedings of the IEEE/CVF Conference on computer vision and pattern recognition (pp. 1521–1530)
30. Simonyan K, Zisserman A (2014) Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*
31. Liu Z, Lin YT, Cao Y, Hu H, Wei YX, Zhang Z, Lin S, Guo BN (2021) Swin transformer: hierarchical vision transformer using shifted windows. In Proceedings of the IEEE/CVF international conference on computer vision (pp. 10012–10022)
32. Ren SQ, Cao XD, Wei YC, Sun J (2014) Face alignment at 3000 fps via regressing local binary features. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 1685–1692)

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